

ICT690 - Special Topic

AI Technologies for Multimedia Applications

Course Project: Art Style Transfer
Qatar painted as Monet's



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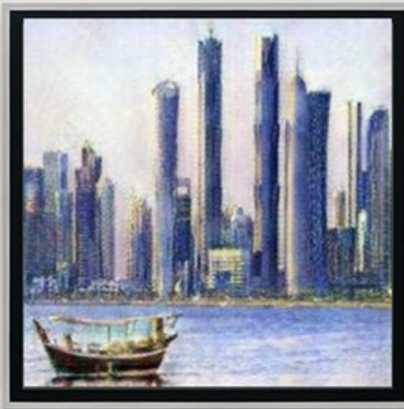
Photo-to-Monet → Qatar-to-Monet: How would Monet render Qatar scenes

Application Industries:

- Art Creation (Qatar Landmarks in Monet Style)
- Video Stylization (Qatar Monet-stylized video)



Museum of
Islamic Art



Doha Corniche



Katara Cultural
village

Relevance to Qatar: cultural goals of the 2030 Qatar National Vision

Nurturing Emerging Talent

This is creating the conditions for creativity and cultivating new talent, inspiring the next generation of cultural producers.

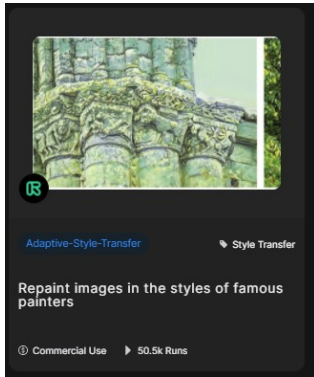
Taking Museums Beyond Four Walls

This is taking the cultural experience beyond the four walls of a museum into the world at large so that more people can take part.

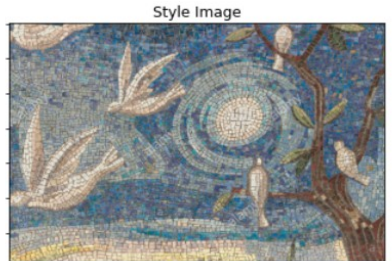
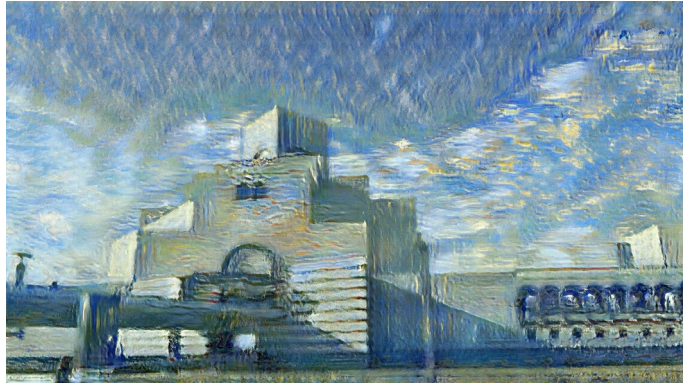
Source: <https://www.qm.org.qa/en/our-purpose>

Related Work: Style Transfer Using GANs

- ☐ Neural Style Transfer (CA12)
- ☐ Unpaired Image-to-Image Translation
- ☐ Adaptive Style Transfer Model



Museum of Islamic Art using Adaptive Style Transfer Model



Museum of Islamic Art using Neural Style Transfer

Claude Monet style

Impressionism: depict lights, shapes, and colours as his eyes perceived them, with **characteristics** such as focus on texture, light and brushstroke

Style:

- ☐ Color not intrinsic to an object, but instead determined by the quality of light thrown on it.
- ☐ Monet also avoided distinctive lines in his painting, instead using dots and patches of color to indicate forms.



Representations of London's parliament by Claude Monet
https://www.ehow.com/info_8350997_major-monet-exemplifies-his-artwork.html



Monet Dataset, examples of brushstrokes

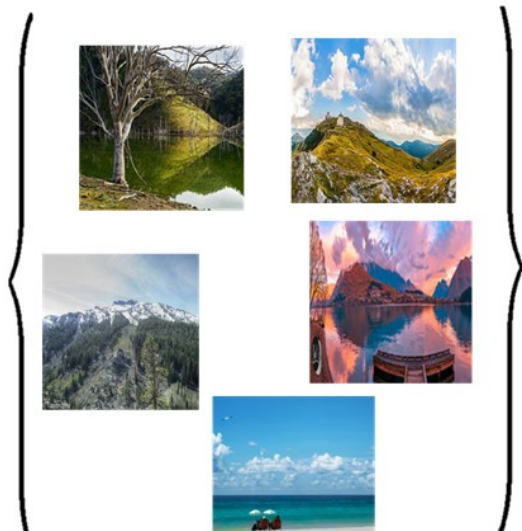
Dataset: Monet2photo

Consist of images from two classes Photos and Monets. (size 256x256)

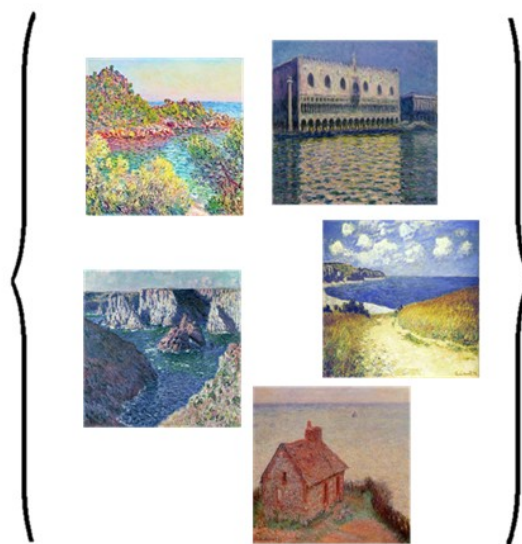
Monet training images: **1072**
Photos training images: **6287**

Monet testing images: **121**
Photos testing images: **751**

Content samples:



Style samples:



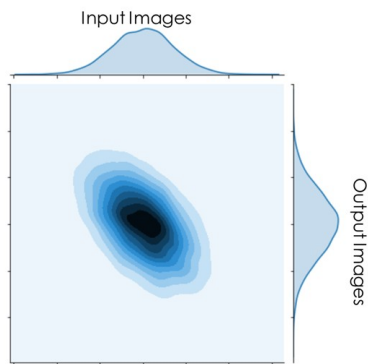
Benchmarking

Qualitative Assessment:

- capture light in color.
- preserving content.

Quantitative Assessment:

Fréchet Inception Distance (FID).



→ FID; the lower the better.

Benchmark comparison with:

1. Baseline:

CycleGAN pix2pix generator

In terms of:

- Training time

2. Pre-trained model:

A Style-Aware Content Loss for Real-time HD Style Transfer

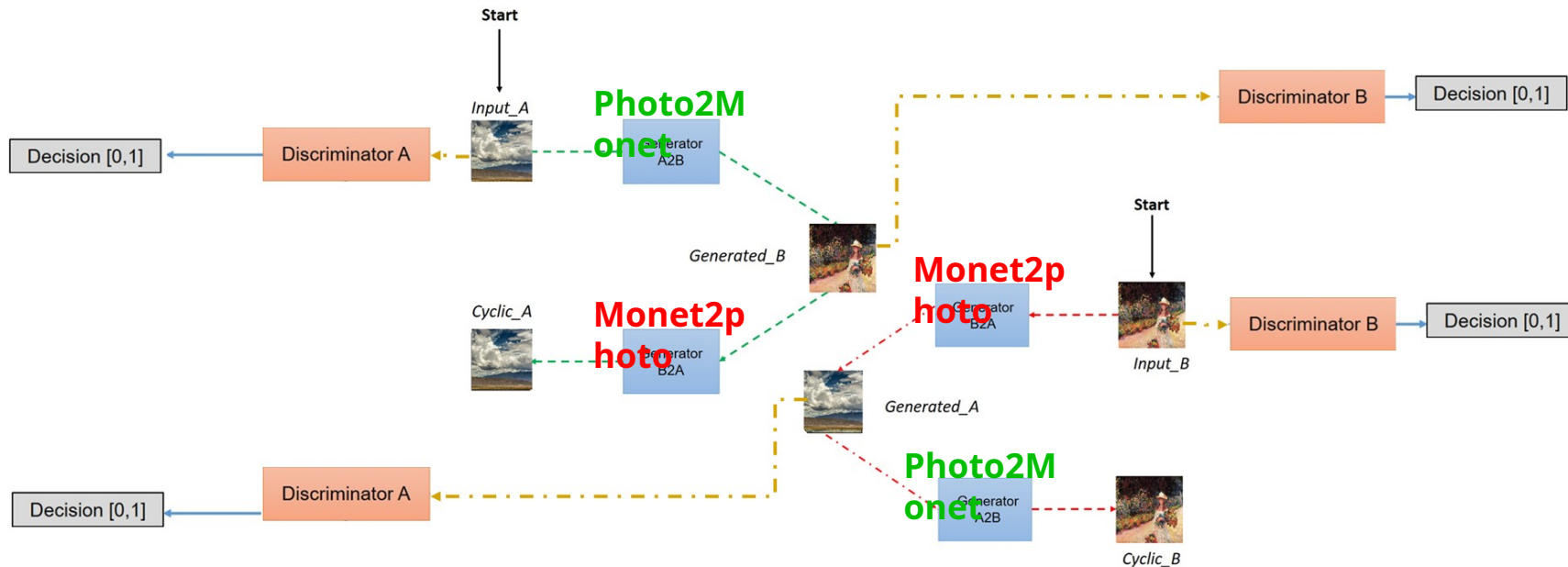
In terms of:

- Inference time

Approach



Network Architecture: CycleGAN

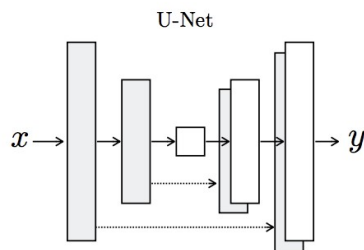


Improvement to baseline:

(CycleGAN paper implementation)

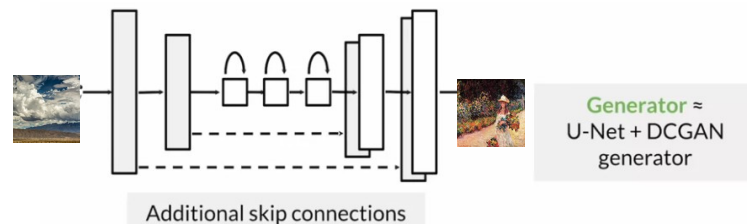
Baseline model generator:

Using U-Net for mapping:



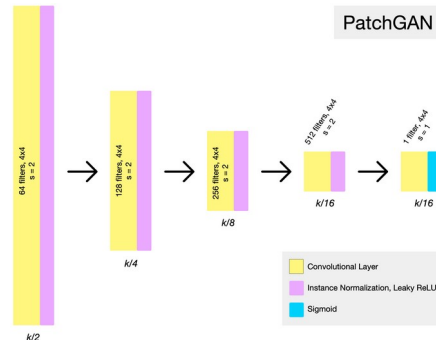
CycleGAN expands up bottleneck section with more convolutional layers from the DCGANs generator, and has additional skip connections.

CycleGAN Generator:



<https://www.coursera.org/learn/apply-generative-adversarial-networks-gans/lecture/euG0f/cyclegan-overview>

CycleGAN Discriminator:

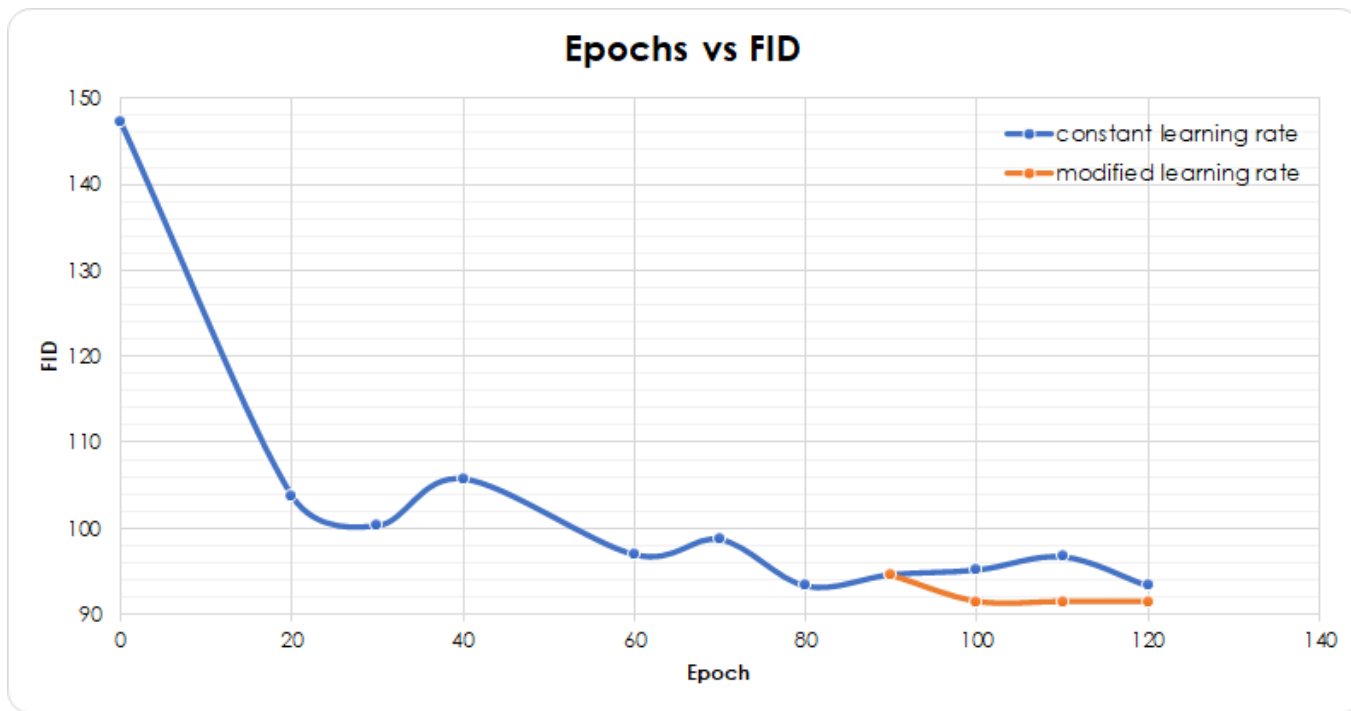


Training: Process



**Manual
Inspection**

Training: Learning Rate Tuning



Benchmarking: Comparison with Baseline

	Pix2Pix GAN	CycleGAN-Res (Our)
Avg. Training Time (Tesla P100)	15 minutes/epoch	7 minutes/epoch
Monet Style: FID (751 test images)	93.587	91.478



Original
|



Pix2Pix GAN



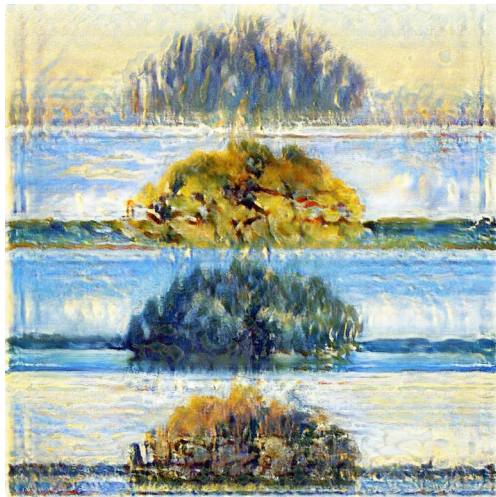
CycleGAN-Res(Our)

Benchmarking: Visual Comparison with Adaptive Style Model

- Capturing light in color & maintaining content



Same Shot different lights
image test

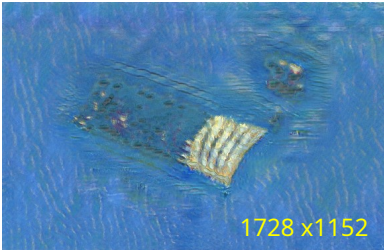


AdaptiveStyle Stylized
Image



Our CycleGAN output

Benchmarking:

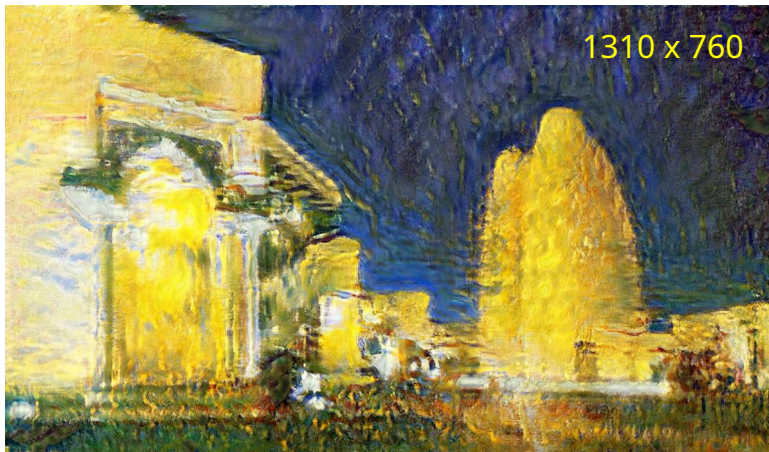
Metric	Adaptive Style	Ours
Inference Time: one image using Colab Tesla T4	84 seconds	0.822 second
Color-preserving  1728 x1152 Qatar National Day Parade 2015	 1728 x1152 Adaptive Style Result	 1728 x1152 Our Stylized Result

Benchmarking:

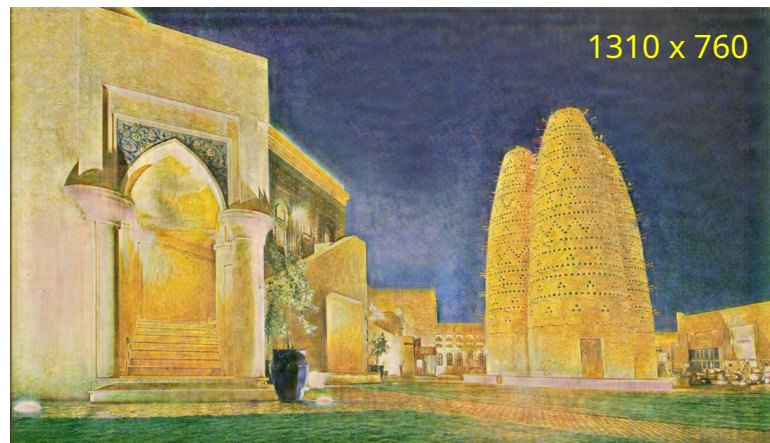
Metric	Image Quality (FID)	Monet Style (FID)
Adaptive Style	313.407	283.793
Ours	137.614	283.957



Katara Cultural Village



Adaptive Stylized Katara Cultural Village



Our Stylized Katara Cultural Village

High Resolution Image Stylization

Input	Logic	Output
High-resolution photo	CycleGAN-Res(ours)	High-resolution stylized image
Low-resolution photo	CycleGAN → Pretrained Image Super Resolution(ISR)	



3,000 x
2,168

Banana Island



3,000 x
2,168

Banana Island Stylized (ours)

High Resolution Image Stylization



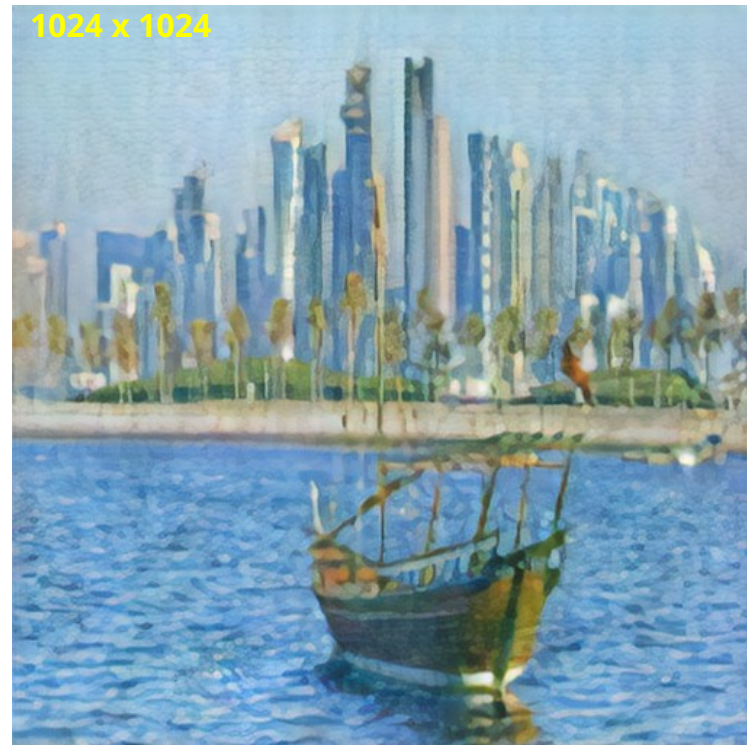
256 x 256

Doha Corniche Low-resolution



256 x 256

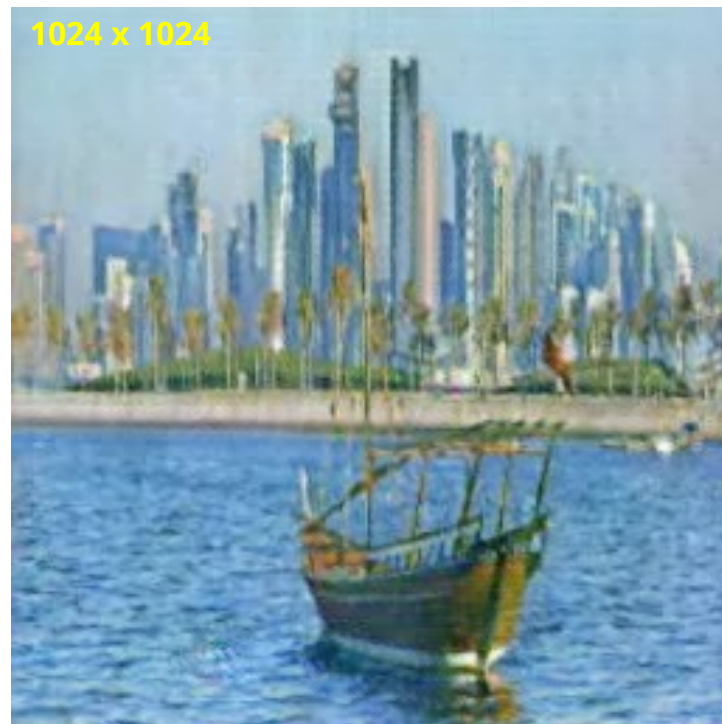
Our CycleGAN Stylized low-resolution

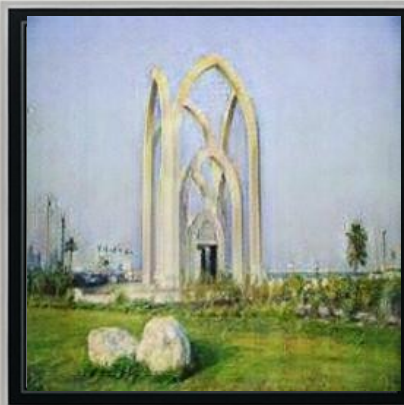


1024 x 1024

Our Stylized Doha Corniche High-resolution with ISP

High Resolution Video Stylization





Thank You,